Full-phase complex Hamiltonian learning with conservative certification and dissipative magnetic feedback

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Anonymous authors

Abstract

Connecting a trained mechanical model to stability and feedback requires error bounds for the function actually deployed. We study this connection for a planar magnetic rod with prescribed configuration-dependent inertia. A two-hidden-layer complex network with 736 real parameters learns a correction depending on both configuration and canonical momentum. A previously fitted 42-parameter positive-semidefinite dissipation head remains fixed during 400 full-batch Adam updates on fourteen COMSOL trajectories; training-only selection returns update 385. A validated 129-dimensional observation design, combined with signed residuals, label fidelity and actual analytic remainders, bounds all first- through third-order derivatives of the frozen energy error in four canonical variables. The same learned Hamiltonian has a separately certified analytic invariant two-torus. Uniform a posteriori bounds also certify an invariant two-torus with a fixed rotation vector for every real member of the conservative learned–reference family, including the prescribed reference endpoint. For dissipative magnetic feedback, power transport retains the learned momentum-gradient error. A physical-reference energy ledger and validated initial-value tubes prove a strict energy cap throughout 200 held commands on the stated finite time horizon. Real/complex and quadratic-prior comparisons assess five paired contrasts. Although the original factorial study favors complex networks, a precision-centered comparison favors the real model; both selected outputs remain less accurate than the shared fourth-order prior on the reused rest-gradient observations. The real dissipation head also performs better in its reported comparison. The results provide deterministic certificates for specified finite-dimensional models; statistical sampling guarantees require their additional assumptions, and a general advantage of complex representations is not established.

Keywords

complex Hamiltonian learning, magnetic soft robots, observation-based error bounds, a posteriori KAM theory, sampled feedback

1 Introduction

Magnetic actuation couples remotely generated fields to magnetized bodies, allowing mechanical input to be delivered without a transmission attached to each moving segment. The resulting motion depends jointly on the actuation field, the body's magnetic properties and its deformation (Abbott et al. 2020). Flexible magnetic robots illustrate how distributed magnetization can support substantial changes of shape and steering in confined environments (Kim et al. 2019). A predictive model is useful for relating an applied command to this evolving configuration. When that model is learned from data, feedback also requires a quantitative account of the effects of its remaining error. A close trajectory fit alone does not specify the domain, time interval or energy margin over which a subsequent stability claim is valid.

Rod and beam mechanics provide a starting point for this analysis. Geometrically nonlinear rod descriptions distinguish the centerline, section orientation, elastic response and applied loads (Antman 2005; Simo 1985). Reduction to a small number of generalized coordinates makes identification and control accessible while preserving selected geometric dependencies. Here a planar magnetic-rod chart provides a finite mechanical reference with elastic energy, magnetic loading and distributed viscous loss. This reference fixes the quantities to which the neural approximation and its certificates refer. The development links mechanical reduction, complex energy learning,

conservative verification and dissipative feedback through one declared canonical model.

An important feature of the reduction is its configuration-dependent inertia. The canonical momentum is obtained from velocity through the positive metric, and the kinetic energy depends on configuration as well as momentum. Its configuration derivative therefore enters the momentum equation. Retaining this dependence provides a consistent conversion between the measured motion, the scalar energy and its canonical field. It also distinguishes a prescribed mechanical prior from the functions that data must determine. We retain the known kinetic graph and a fourth-order potential, then allow a neural correction to depend on all four canonical coordinates. This permits momentum-dependent model error to enter both identification and verification explicitly.

Hamiltonian neural networks learn a scalar energy and differentiate it to obtain a Hamiltonian vector field (Greydanus et al. 2019). The scalar representation links the components of the predicted dynamics and imposes the conservative differential structure algebraically. A mechanical prior can further specify an inertia or an elastic component while leaving a smaller correction to be learned. For the present full-phase correction, this construction retains a real learned energy even though the hidden features and weights are complex. The learned velocity is the momentum derivative of that complete scalar. It contains the network's momentum derivative in addition to the velocity associated with the prescribed kinetic energy.

Dissipation and input have complementary roles. Dissipative SymODEN encodes a port-Hamiltonian computation graph containing energy loss and control (Zhong et al. 2020). Port-Hamiltonian neural networks also separate a stationary energy from dissipative and driving terms in learning non-autonomous dynamics (Desai et al. 2021). We use a configuration-dependent factor decoder for a real positive-semidefinite dissipation matrix. Its quadratic Rayleigh function gives a dissipative force through a velocity derivative. The canonical decoder combines that force with the full learned Hamiltonian velocity and an explicit magnetic port. This construction yields a checkable power identity and specifies how a separately fitted matrix head can be paired with the conservative energy. The pair's mathematical identity is maintained when its errors enter a reference-plant controller.

Complex networks offer a concrete parameterization for these scalar and matrix functions. Trabelsi et al. (2018) develop building blocks for deep complex learning, including complex affine operations and their real implementation. The conservative model used here has two literal complex-tanh hidden layers, independent affine coefficients for the complex input and its conjugate on real states, and a real scalar readout. Evenization and an energy gauge fix its symmetry and value at the origin. For analytic verification, the four real canonical coordinates are extended independently to complex arguments. The real readout then has a coefficient-conjugate analytic continuation. Avoidance of the tanh poles becomes a candidate-specific mathematical condition on that continuation.

The use of complex arithmetic also raises an empirical question about representation. Zhang et al. (2022) study approximation and optimization properties of specified complex-reaction network families. Those results motivate examining network structure and training conditions when comparing real and complex models. Our finite-width complex-tanh model, its constrained parameterization and its Adam objective are evaluated through their actual returned functions. We compare real and complex architectures both with and without a prescribed quadratic Hamiltonian, and separately examine the effect of objective and output scaling. This separates the practical choice of a representation from the canonical and power identities, which follow from the scalar and decoder construction.

Nonzero approximation error is central to the stability question. Chen et al. (2022) connect Hamiltonian neural networks with nonzero training loss to KAM theory through statistical learning and regularity estimates. Their analysis makes the sampling distribution and control of the relevant function class substantive ingredients. Classical KAM theory describes persistence of suitable quasiperiodic invariant tori under small Hamiltonian perturbations, with arithmetic and nondegeneracy conditions (Pöschel 2001). For a computed mechanical model, an a posteriori theorem starts from a proposed torus and checks its invariance defect, analytic domain and geometry (Haro and Luque 2019). Learning error can contribute to these estimates, but the target torus and its complete defect remain additional objects to be verified.

Trajectory observations constrain a different set of quantities from those analytic hypotheses. Successive observations share prescribed motions and initial conditions. Weak momentum balances constrain integrals along those motions, whereas an analytic stability proof needs derivatives on a whole state domain. Configuration-gradient observations at zero momentum give useful potential information but do not determine arbitrary momentum dependence. We therefore combine rest-gradient information with momentum-gradient residuals at observation-chart anchors. Because the kinetic

graph is shared, the reference correction has an exactly zero momentum derivative at those anchors. This structural identity provides a full-phase observation channel without using exported native momentum derivatives or reference forces as training labels.

The realized observation design is verified through finite-dimensional representatives of the actual network output. For the degree-six representative, all nonconstant even-total-degree monomials in the four canonical variables give 129 coordinates. Fifteen configuration-only coordinates and 114 mixed coordinates are recovered with validated inverse bounds. The neural function remains the deployed function; the representative serves to transport its signed observation residual and its actual remainder. Observation-point tails, target-domain tails, label fidelity and the inverse correction enter the same inequality. This yields complete mixed-derivative bounds rather than inferring full-phase accuracy from a configuration slice.

A prospective sampling result complements this realized-design calculation. Independent anchors under a specified importance law can give a common conditioning event for the derivative design. On that event, every candidate with sound individual residual and remainder certificates satisfies the corresponding output-error implication. Consequently an output may be selected using the same observations without a union bound over all returned network parameters. The probability statement controls accepting an output whose asserted guarantee is false. Its assumptions remain explicit, including the independent sampling law and common fidelity event. The prescribed fourteen trajectories support the deterministic certificate used here.

Time discretization supplies another structural component. Symplectic integration preserves the symplectic form of a conservative Hamiltonian flow; implicit midpoint is a standard example (Hairer 2010). We formulate that method for the nonseparable mechanical energy and its complex-coordinate representation. Local solvability and global approximation estimates retain the contribution of a finite nonlinear-solver residual. The numerical approximation, continuous-flow KAM calculation and recorded COMSOL observations thus have their own specified roles. This allows the discretization analysis to support the same mechanical definitions without substituting a discrete-map property for a continuous invariant-torus certificate.

The principal trained instance is a 736-real-coordinate complex energy correction with an already fitted 42-coordinate PSD matrix head held fixed. It uses fourteen distributed-viscosity COMSOL trajectories and 400 full-batch Adam updates. Selection by the training objective gives update 385 before the reused rest labels are evaluated. Its complete four-dimensional observation-error certificate bounds every mixed derivative through order three on the stated complex rectangle. Certification uses this precision-centered GC checkpoint; the three factorial GC outputs retain their own parameter identities. A separate computer-assisted calculation proves an analytic invariant two-torus for this same learned Hamiltonian. The observation-derived error bounds also enter a completed conservative-family certificate. Uniform domains, the actual varying mean torsion and a complete whole-strip defect for a degree-ten response satisfy all eight a posteriori inequalities. For each real family parameter, these bounds certify a real analytic embedded torus with the same exact rotation vector, including at the prescribed conservative reference endpoint.

For magnetic feedback, the difference between the learned and reference velocities contributes to both the magnetic work allowance and the dissipation credit. We retain this momentum-gradient error when constructing the held-command filter. A rational energy ledger and validated initial-value tubes then establish a strict physical-reference energy cap throughout 200 holds on the declared finite horizon. The result concerns the specified numerical mechanical reference and distributed viscosity. It supplies an entire-hold interpretation of the sampled decisions while keeping reference calibration and hardware interfaces outside that numerical assertion.

The comparisons also expose the limits of the learned correction. The real/complex by quadratic-prior experiment retains five paired contrasts over three common-data optimization seeds. It favors the complex architecture under its original budget, but its absolute rest-gradient errors exceed the shared fourth-order prior's error. A precision-centered prior-assisted comparison reverses the real–complex ranking: the real output is more accurate, and both selected outputs remain worse than that prior on the reused labels. The real dissipation head also performs better on the reported matrix identification metrics. These results make initialization, scaling, architecture and the physical prior part of the interpretation of a representation comparison.

The contribution is a quantitative connection between an actually trained full-phase complex energy, its observation-error certificate, the conservative learned–reference family's invariant tori and a separate dissipative physical-reference safety certificate. The analysis keeps the network, canonical chart, finite mechanical graph and decoder consistent across these stages. The reported family certificate uses realized deterministic observations; its statistical extensions retain the

additional sampling and class-wide regularity hypotheses. The resulting framework allows structural validity, approximation accuracy and downstream mathematical acceptance to be assessed for the same returned functions.

2 Mechanical reduction and complex symplectic learning

2.1 Reduced rod mechanics and physical normalization

The physical reference is a planar, inextensible and unshearable Kirchhoff rod with a fixed base and a magnetic tip moment. Two prescribed shape functions supply the dimensionless coordinates $q \in \mathbb{R}^2$. Writing $\beta_i(s) = f_i'(s)$, its angular and centreline descriptions are

$$\theta(s,q) = \beta(s)^{\mathsf{T}} q, \qquad r(s,q) = \int_0^s \begin{pmatrix} \sin\theta(\xi,q) \\ \cos\theta(\xi,q) \end{pmatrix} \mathrm{d}\xi. \tag{1}$$

The spatial components are transverse x and axial z. The trigonometric tangent retains finite rotations, while the prescribed two-coordinate chart restricts the admissible shapes. No axial stretching or independent shear variable is added. The construction follows rod kinematics Antman (2005); Simo (1985); all conclusions below concern its specified finite reduction.

The cross-section follows the local tangent, so its angular velocity is linear in the coordinate velocities even when the centreline undergoes finite rotation. This accounts for the constant rotary contribution below. The bending constitutive law remains quadratic in curvature within this chart. Gravity, field gradients and additional body-force terms are absent from the reference equations. The free end receives the magnetic moment; its rotation is not prescribed as an essential boundary condition.

2.1.1 The fixed quadrature and inertia The shape functions are piecewise cubic on 32 cells of the 0.08 m rod. At a material location, the configuration Jacobian is

$$J(s,q) = \partial_q r(s,q) = \int_0^s \begin{pmatrix} \cos\theta(\xi,q) \\ -\sin\theta(\xi,q) \end{pmatrix} \beta(\xi)^{\mathsf{T}} \,\mathrm{d}\xi. \tag{2}$$

The implemented $J_{ek}(q)$ uses eight-point Gauss quadrature: completed cells contribute their full outer sums, and the current cell contributes a nested eight-point sum up to its outer node. With cell length ℓ_e and positive stored weights w_k, define

$$\begin{aligned} D(q) &= \sum_{e,k} \ell_e w_k J_{ek}(q)^{\mathsf{T}} J_{ek}(q), \\ G(q) &= f_m D(q) + K_r = S(q) S(q)^{\mathsf{T}}. \end{aligned} \tag{3}$$

Here K_r is the constant normalized rotary-inertia matrix, and $f_m > 0$ is the stored translational factor, nominally $\omega_0^2 \rho A_c / E_0$. In the mechanical and dissipation formulas, S denotes the lower Cholesky factor, not a further modal scaling. The known G is positive definite on the certified configuration box $|q_i| \le 1/10$, and S has strictly positive diagonal. The Gauss nodes, weights and mechanical coefficients are frozen binary64 values. Their exact stored values define the analytic reference; freshly evaluating a continuum integral or a parameter product would define a different reference. In particular, the stored f_m is retained in all identities involving $G - K_r$.

Translational kinetic energy is assembled from the velocities of material points, using their configuration Jacobians before the Gram products are formed. Consequently its inertia depends on configuration even though the two shape functions themselves are fixed. The same metric is used when constructing native trajectories, converting observed velocity to momentum, evaluating the learned Hamiltonian and verifying its conservative flow. The required derivatives differentiate this configuration dependence.

2.1.2 Canonical energy and dimensional quantities Let t denote SI time and set $\tau = \omega_0 t$, with $\omega_0 \simeq 11.417287282852458\,\mathrm{s}^{-1}$ and $E_0 = \mu B_0 \simeq 6.0 \times 10^{-7}\,\mathrm{J}$. Here μ is the tip magnetic dipole magnitude and B_0 the constant bias field. These scales and the shape functions are fixed before learning. A prime denotes $\mathrm{d}/\mathrm{d}\tau$, so $v = q' = \dot{q}/\omega_0$. Dividing physical energy by E_0 gives

$$\begin{aligned} p &= G(q) v, \\ H_*(q,p) &= \tfrac{1}{2} p^{\mathsf{T}} G(q)^{-1} p + V_*(q), \\ V_*(q) &= \tfrac{1}{2} q^{\mathsf{T}} B q + 1 - \cos(d^{\mathsf{T}} q). \end{aligned} \tag{4}$$

The fixed $B \succ 0$ is the normalized bending matrix, and the fixed vector d maps the chosen coordinates to tip rotation. Both are the exported coefficients of this chart. Thus the full bias-field cosine and the configuration-dependent inertia are retained. The momentum p is dimensionless canonical momentum, not SI velocity; if P_{SI} is the momentum conjugate to the same dimensionless q, then $p = (\omega_0/E_0) P_{\mathrm{SI}}$.

Equivalently, the dimensionless Lagrangian is $L_*(q,v) = v^{\mathsf{T}} G(q) v/2 - V_*(q)$, and the momentum definition in (4)

is its Legendre transform. In particular, configuration derivatives of the kinetic energy contribute to the momentum equation. They cannot be replaced by the force of a constant-mass oscillator. The prescribed mechanical graph defines both the numerical reference and the observation chart. The learned scalar adds a correction on the full position–momentum space. Its momentum derivative changes the learned velocity; the reference identity $v = G^{-1}p$ is not imposed on that learned velocity.

Distributed viscous drag supplies the physical Rayleigh coefficient $C_{\mathrm{phys}} = c_{\mathrm{line}} D(q)$, whose units are J s: J has units m, D has units m^3, and c_{line} has units $\mathrm{N\,s/m^2}$. Its normalized damping is

$$\begin{aligned} R_*(q) &= \frac{\omega_0 c_{\mathrm{line}}}{E_0} D(q) \\ &= a\{G(q) - K_r\}, \\ a &= \frac{\omega_0 c_{\mathrm{line}}}{E_0 f_m}, \\ \mathcal{D}_*(q,v) &= \tfrac{1}{2} v^{\mathsf{T}} R_*(q) v. \end{aligned} \tag{5}$$

The declared $c_{\mathrm{line}} = 0.0006911873093836746\,\mathrm{N\,s/m^2}$ is a numerical-study assumption inherited from a nominal small-slope damping estimate, not a measured material calibration. The present reference sets tip drag $c_{\mathrm{tip}} = 0$ and additional mass-proportional drag $\gamma_{\mathrm{add}} = 0$. Its damping is the matrix in (5); no constant scalar-drag law is substituted.

2.1.3 Magnetic input and the power balance A transverse field B_y gives the normalized input $u = B_y/B_0$ and interaction potential $-u\sin(d^{\mathsf{T}}q)$. Treating this interaction as external work yields the single physical port

$$\begin{aligned} g(q) &= d\cos(d^{\mathsf{T}}q), \\ y &= g(q)^{\mathsf{T}} v, \\ q' &= \partial_p H_*, \\ p' &= -\partial_q H_* - R_* v + g(q)u. \end{aligned} \tag{6}$$

Consequently $H_*' = -v^{\mathsf{T}} R_* v + uy$ and the SI power is $E_0\omega_0 H_*' = -\dot{q}^{\mathsf{T}} C_{\mathrm{phys}} \dot{q} + \mu B_y \cos(d^{\mathsf{T}}q) d^{\mathsf{T}}\dot{q}$. The distributed-viscosity identification trajectories use $u = 0$; the forced validation and feedback runs apply prescribed or held inputs through this same port. COMSOL Global Equations integrates these four reduced canonical equations in SI time. It does not provide a three-dimensional finite-element rod solution or a PDE-convergence certificate. When a conservative KAM statement is invoked, it concerns the subsystem obtained by setting both the input and dissipation to zero while retaining the same metric and stored potential. The dissipative training reference and the forced control system have their own power and stability statements; conservation is not inferred for either of those flows.

2.2 Full-phase complex energy and dissipative learning

The prescribed kinetic graph is $T_G(q,p) = p^{\mathsf{T}} G(q)^{-1} p/2$, with the positive metric G obtained from the stated finite Gauss reduction. Write

$$\begin{aligned} H_4 &= T_G + V_4(q), \\ \widehat{H}_\vartheta &= H_4 + N_\vartheta(q,p), \\ H_* &= H_4 + U(q). \end{aligned} \tag{7}$$

Here V_4 includes the prescribed quadratic and quartic potential terms. The correction $U(q) = 1 - \cos(s) - s^2/2 + s^4/24$, with $s = d^{\mathsf{T}}q$, defines the conservative reference remainder. It is used in verification; its coefficients are not training labels. The learned correction depends on all four canonical coordinates. Consequently the deployed velocity is

$$\widehat{v} = \partial_p \widehat{H}_\vartheta = G^{-1}p + \partial_p N_\vartheta. \tag{8}$$

2.2.1 The scalar network and its analytic continuation Let $s_j > 0$ be the stored physical input scales, and put $y_j = x_j/s_j$ for $x = (q_1, q_2, p_1, p_2)$. The first layer receives the four complex features

$$\begin{aligned} z_j^+ &= (y_j + i y_{2+j})/s_0, \\ z_j^- &= (y_j - i y_{2+j})/s_0, \qquad j = 1, 2, \end{aligned}$$

where s_0 is the stored binary64 value of $\sqrt{2}$. On real states the two feature pairs are conjugates. Their coefficients are not tied by conjugacy; their effective values obey the declared raw-parameter constraints. Two width-sixteen complex affine–tanh layers are followed by a bias-free complex linear readout f_ϑ. The model has 736 trainable real coordinates. The real scalar correction is

$$N_\vartheta(x) = \eta \left\{ \frac{\operatorname{Re} f_\vartheta(y) + \operatorname{Re} f_\vartheta(-y)}{2} - \operatorname{Re} f_\vartheta(0) \right\}. \tag{9}$$

For this fit η is the stored energy scale multiplied by 2^{-28}. Evenization and the gauge imply $N_\vartheta(-x) = N_\vartheta(x)$ and $N_\vartheta(0) = 0$. There is no monomial prefactor or separately fitted polynomial replacing the hidden layers.

For complex-domain analysis, $x_1, \ldots, x_4$ are independent complex variables. The displayed linear expressions define $z^{\pm}$ there; they are not evaluated by conjugating a variable point. If $f_\vartheta^\sharp(y) = \overline{f_\vartheta(\overline{y})}$ denotes coefficient conjugation, replace each real readout in (9) by $(f_\vartheta + f_\vartheta^\sharp)/2$. This gives a holomorphic continuation wherever the actual tanh

preactivations avoid their poles. Pole separation is certified for the frozen output on each required domain. The defining layer constraints, input buffers and pre-rounded smooth-absolute constants are reconstructed from the raw checkpoint; rounded exported effective weights do not define a second model.

2.2.2 A separate positive-semidefinite dissipation head

A configuration-only complex network with two width-two tanh layers and 42 raw real coordinates produces a lower triangular factor $L_\psi(q)$. With $G = SS^{\mathsf{T}}$, define

$$\begin{aligned} \widehat{M}_\psi &= \mathrm{Re}(L_\psi L_\psi^*), \\ \widehat{R}_\psi &= S\widehat{M}_\psi S^{\mathsf{T}}, \\ \widehat{\mathcal{D}}_\psi(q,v) &= \tfrac{1}{2} v^{\mathsf{T}} \widehat{R}_\psi(q) v. \end{aligned} \tag{10}$$

For real w, $w^{\mathsf{T}}\widehat{M}_\psi w = \|L_\psi^* w\|_2^2 \geq 0$, and $\partial_v \widehat{\mathcal{D}}_\psi = \widehat{R}_\psi v$. The deployed canonical decoder is therefore

$$\begin{aligned} q' &= \widehat{v}, \\ p' &= -\partial_q \widehat{H}_\vartheta - \widehat{R}_\psi \widehat{v} + g(q)u, \\ g(q) &= d\cos(d^{\mathsf{T}} q). \end{aligned} \tag{11}$$

The pair $\widehat{H}_\vartheta + i\widehat{\mathcal{D}}_\psi$ records an energy and a Rayleigh function. The equations are defined by (11), rather than by treating the pair as a single holomorphic Hamiltonian generator. In particular,

$$\widehat{H}'_\vartheta = -\widehat{v}^{\mathsf{T}} \widehat{R}_\psi \widehat{v} + u\, g^{\mathsf{T}} \widehat{v}.$$

Both occurrences of $\widehat{v}$ retain $N_{\vartheta,p}$.

2.2.3 The observed weak balance and the executed objective

The conservative fit uses fourteen native COMSOL trajectories through time, configuration, observed velocity and prescribed input. The observation chart sets $\Pi = G(q)v$; exported native momenta, momentum derivatives and reference damping values are not optimizer labels. For window w, let Q_w be its declared positive quadrature, h_w its duration, and define

$$\begin{aligned} r_{4,w} &= D_{\dot p}^{-1} \frac{\Delta\Pi_w - Q_w[-H_{4,q}(q,\Pi) - \widehat{R}_\psi v + g(q)u]}{h_w}, \\ r_{\vartheta,w} &= r_{4,w} + D_{\dot p}^{-1} \frac{Q_w[N_{\vartheta,q} + \widehat{R}_\psi N_{\vartheta,p}]}{h_w}. \end{aligned} \tag{12}$$

All network derivatives here are evaluated at (q,Π), and $D_{\dot p}$ contains the fixed positive momentum-rate scales. The signs follow by integrating the momentum equation in (11). Centering evaluates the perturbation directly and retains the actual scalar H_4 baseline.

Let n_w be the number of windows and let D_v contain the fixed positive kinematic scales. With the same positive quadrature, the executed full-batch objective is

$$\begin{aligned} F(\vartheta) = \frac{1}{F_4}\Bigg\{ &\frac{1}{2n_w}\sum_w \|r_{\vartheta,w}\|_2^2 \\ &+ \frac{0.1}{2n_w}\sum_w \frac{Q_w[\|D_v^{-1} N_{\vartheta,p}\|_2^2]}{h_w} \Bigg\}, \end{aligned} \tag{13}$$

where $F_4 > 0$ is the fixed initial weak-loss value. The kinematic residual is $N_{\vartheta,p}$ because $H_{4,p}(q,\Pi) = v$ in the prescribed observation chart. No raw-parameter penalty is included in this fit.

The matrix head is already frozen when these conservative coordinates are trained. The two networks are composed sequentially. Full-batch Adam uses 400 updates, learning rate 0.01, and $\epsilon = 10^{-8}$. Every network coordinate remains trainable. The initially zero readout gives the exact initial scalar H_4; hidden-layer changes with a nonzero functional effect are recorded after the second update. Training-only minimization over the initial and recorded objective values selects update 385. The rest-label cohort is used subsequently for reused validation and finite-output certification. It does not enter that checkpoint selection.

The error analysis applies to this stored output and its actual observation residuals. It does not require that Adam has reached a stationary point, nor does it substitute a convergence theorem for the executed optimization. The full-phase real comparison and the earlier factorial study retain their own training budgets and outputs.

2.3 Complex canonical form and symplectic discretization

The complex canonical coordinates are derived from the same full-phase scalar energy as the real equations. Their exact coordinate change is distinct from the stored numerical preprocessing inside the network. The conservative midpoint method is classical (Hairer 2010, Lecture 2, Theorem 3); the formulas below retain both the configuration-dependent inertia and the network's momentum derivatives.

2.3.1 The complex equation of the actual scalar energy

For normalized time τ, set

$$\begin{aligned} z &= (q + ip)/\sqrt{2}, \quad \bar z = (q - ip)/\sqrt{2}, \\ q &= (z + \bar z)/\sqrt{2}, \quad p = (z - \bar z)/(i\sqrt{2}). \end{aligned} \tag{14}$$

The square root here is the exact mathematical value; the network's fixed input buffer s_0 remains part of its own scalar graph. Put $\mathcal{H}(z,\bar z) = \widehat{H}(q(z,\bar z), p(z,\bar z))$. Treating z and $\bar z$

independently in the chain rule gives

$$\partial_z \mathcal{H} = (\widehat{H}_q - i\widehat{H}_p)/\sqrt{2}, \qquad \partial_{\bar z} \mathcal{H} = (\widehat{H}_q + i\widehat{H}_p)/\sqrt{2}. \tag{15}$$

Therefore the real canonical equations are equivalent to

$$z' = (\widehat{H}_p - i\widehat{H}_q)/\sqrt{2} = -i\partial_{\bar z}\mathcal{H}. \tag{16}$$

The corresponding real symplectic form is $\sum_j dq_j \wedge dp_j = i\sum_j dz_j \wedge d\bar z_j$.

Write $W(q) = G(q)^{-1}$, $v_* = W(q)p$ and $G_j = \partial_j G$. Differentiating $WG = I$ gives $\partial_j W = -WG_jW$. For the deployed scalar $\widehat{H} = p^{\mathsf T}Wp/2 + V_4 + N(q,p)$, its complete derivatives are

$$\widehat{H}_p = v_* + N_p, \qquad \widehat{H}_{q_j} = -\tfrac{1}{2}v_*^{\mathsf T}G_j v_* + \partial_j V_4 + N_{q_j}. \tag{17}$$

Thus the configuration derivative of inertia and the full neural gradient enter (16). In particular, v_* is the reference velocity; it is not the full learned velocity.

The physical real scalar generally depends on both z and $\bar z$. Its analytic continuation instead extends q, p to independent complex coordinates Q, P. The two linear combinations $(Q + iP)/\sqrt{2}$ and $(Q - iP)/\sqrt{2}$ are then independent coordinates. For the actual network, the continuation $N^{\mathbb C}$ uses coefficient-sharp readouts as defined in Section 2.2, giving

$$\widehat{H}^{\mathbb C}(Q,P) = \tfrac{1}{2}P^{\mathsf T}G(Q)^{-1}P + V_4(Q) + N^{\mathbb C}(Q,P). \tag{18}$$

The domain must exclude singularities of G^{-1} and the literal tanh activations. Matrix transposes in this continuation are ordinary analytic transposes. Taking the real part of a variable complex evaluation would not define the required holomorphic continuation.

The configuration-only Rayleigh head satisfies $\partial_v \widehat{\mathcal{D}} = \widehat{R}(q)v$. Its canonical decoder evaluates this derivative at $v = \widehat{H}_p$. Adding the magnetic port therefore yields

$$z' = -i\partial_{\bar z}\mathcal{H} + \frac{i}{\sqrt{2}}\{-\widehat{R}(q)\widehat{H}_p + ug(q)\}. \tag{19}$$

The time discretization below concerns the conservative subsystem obtained by setting the Rayleigh force and input to zero.

2.3.2 Implicit midpoint with all momentum derivatives retained Let $x = (q,p)$, $J_c = \left(\begin{smallmatrix} 0 & I_2 \\ -I_2 & 0 \end{smallmatrix}\right)$ and $f = J_c\nabla\widehat{H}$. A midpoint evaluation defines

$$x_{n+1} = x_n + hJ_c\nabla\widehat{H}(m_n), \qquad m_n = (x_n + x_{n+1})/2. \tag{20}$$

Writing $\bar q = (q_n + q_{n+1})/2$, $\bar p = (p_n + p_{n+1})/2$ and $\bar v_* = G(\bar q)^{-1}\bar p$ gives the actual coupled equations

$$\begin{aligned} q_{n+1} &= q_n + h\{\bar v_* + N_p(\bar q, \bar p)\},\\ p_{n+1,j} &= p_{n,j} + h\{\tfrac{1}{2}\bar v_*^{\mathsf T}G_j(\bar q)\bar v_* - \partial_j V_4(\bar q) - N_{q_j}(\bar q,\bar p)\}. \end{aligned} \tag{21}$$

Equivalently, (14) gives the complex midpoint step

$$z_{n+1} = z_n - ih\partial_{\bar z}\mathcal{H}((z_n + z_{n+1})/2, (\bar z_n + \bar z_{n+1})/2).$$

For a closed ball $\overline{B}(x,b)$ in a real domain with $\widehat{H} \in C^3$, suppose $\|f\| \le M$ and $\|Df\| \le L$. If

$$|h|M < b, \qquad |h|L/2 < 1, \tag{22}$$

then $y \mapsto x + hf((x+y)/2)$ maps that ball into itself and is a contraction. Its unique fixed point defines a differentiable local map Φ_h by the implicit-function theorem.

Proposition 1. Symplecticity of the conservative midpoint map. *Under* (22)*, the exactly solved local step satisfies* $(D\Phi_h)^{\mathsf T}J_cD\Phi_h = J_c$.

Proof. At its midpoint put $A = J_c\nabla^2\widehat{H}(m_n)$. Symmetry of the full scalar Hessian gives $A^{\mathsf T}J_c + J_cA = 0$. Differentiation of (20) yields

$$(I - hA/2)D\Phi_h = I + hA/2.$$

The matrices $U_h = I - hA/2$ and $V_h = I + hA/2$ commute, and

$$V_h^{\mathsf T}J_cV_h = J_c + (h^2/4)A^{\mathsf T}J_cA = U_h^{\mathsf T}J_cU_h.$$

Since $D\Phi_h = V_hU_h^{-1}$, multiplication on the left by $U_h^{-\mathsf T}$ and on the right by U_h^{-1} proves the claim. No separability or vanishing momentum derivative of N was used.

2.3.3 Numerical error and its distinction from learning error Use the maximum-component norm in physical canonical coordinates, its induced matrix norm and the corresponding bilinear norm. Let a common convex neighborhood contain all exact and numerical states and segments under comparison, and let M, L, B_2 bound f, Df, D^2f in these norms. The observational field bound δ_0 below uses this same physical-coordinate maximum norm. Choose $h_0 > 0$ with $h_0L/2 = \kappa < 1$, require $0 < h \le h_0$, and put

$$C_d = \frac{B_2M^2}{24} + \frac{L^2M}{6}, \qquad C_\ell = \frac{C_d}{1-\kappa}, \qquad \Lambda = \frac{L}{1-\kappa}.$$

The exact endpoint's midpoint-equation defect is at most $C_d h^3$. Contraction then bounds the one-step state error by $C_\ell h^3$. The step's Lipschitz factor is at most $(1 + hL/2)/(1 - hL/2) \le e^{\Lambda h}$, so for equal initial states and $nh \le T$,

$$\|\Phi_h^n(x_0) - \phi_{\widehat{H}}^{nh}(x_0)\| \le C_\ell h^2 \frac{e^{\Lambda T} - 1}{\Lambda}. \tag{23}$$

The quotient is T when $\Lambda = 0$. Supplementary Section S1 proves these constants by integral remainders and includes finite nonlinear-solver residuals. Such residuals need not preserve the exact symplectic identity.

By comparison, if $\|X_{\widehat{H}} - X_{H_*}\| \le \delta_0$ and X_{H_*} has Lipschitz bound L_* on a common trajectory neighborhood, the equal-initial-state conservative flows satisfy

$$\|\phi_{\widehat{H}}^t(x_0) - \phi_{H_*}^t(x_0)\| \le \delta_0 \frac{e^{L_* t} - 1}{L_*}. \tag{24}$$

This follows from the integral difference equation and Gronwall's inequality, with quotient t when $L_* = 0$. Its error term is the actual learned-field discrepancy, which is bounded from observations below. The numerical and learned-model errors therefore enter separate terms; a small time step cannot remove an inaccurate learned field.

For a quadratic energy the exact midpoint rule preserves that energy, but the deployed nonquadratic scalar generally has an additional energy increment. The full learned dissipative vector field has canonical divergence

$$-\operatorname{tr}(\widehat{R}\widehat{H}_{pp}) + g(q)^{\mathsf{T}} \nabla_p u.$$

For prescribed input the second term vanishes. The first uses the actual Hessian $\widehat{H}_{pp} = G^{-1} + N_{pp}$, rather than replacing it by the reference inverse metric. For prescribed input, a strictly positive trace implies volume contraction and excludes symplecticity of that dissipative flow. With state-dependent feedback, the corresponding sufficient condition is $\operatorname{tr}(\widehat{R}\widehat{H}_{pp}) - g(q)^{\mathsf{T}} \nabla_p u > 0$. The certified KAM conclusions concern the continuous conservative flow; a discrete-map KAM statement would require its own step-size nonresonance, map geometry and invariance-defect certificate.

3 Numerical approximation and observation-based learning error

3.1 Comparison protocols and the certified output

The four-cell factorial and the later precision-centered comparison use related architectures but different objectives, output scales and checkpoints. The labels R, C, GR and GC in the factorial tables refer only to that protocol. The deployed GC in (7) is the selected output of the precision-centered comparison. Function-specific error, KAM and feedback statements below always refer to that latter output.

3.2 Generic phase-space networks: representation and quadratic prior

A factorial experiment compares real and complex networks with and without a fixed quadratic Hamiltonian. It uses the current finite-Gauss mechanics and distributed-viscosity observations. The four cells are R (real, no fixed quadratic term), C (complex, no fixed quadratic term), GR (real with the quadratic prior), and GC (complex with the quadratic prior). The quadratic prior improves both paired representations, and GC has the smallest factorial error for every seed. The absolute errors nevertheless remain substantially larger than those of the shared fourth-order model with the network correction set to zero.

Writing $G_0 = G(0)$ and $\mathsf{K}_2 = B + dd^T$, every cell has the form

$$\begin{aligned} H_s(q,p) &= T_{\mathrm{nl}}(q,p) + V_{\mathrm{quartic}}(q) \\ &\quad + I_s H_2(q,p) + N_{\theta,s}(q,p), \\ T_{\mathrm{nl}}(q,p) &= \tfrac{1}{2} p^T [G(q)^{-1} - G_0^{-1}] p, \\ V_{\mathrm{quartic}}(q) &= -\tfrac{1}{24} (d^T q)^4, \\ H_2(q,p) &= \tfrac{1}{2} p^T G_0^{-1} p + \tfrac{1}{2} q^T \mathsf{K}_2 q, \end{aligned} \tag{25}$$

where $I_s = 0$ for R and C and $I_s = 1$ for GR and GC. Thus the ablation removes the complete origin quadratic Hamiltonian, including its momentum block. The known nonlinear kinetic remainder and quartic magnetic term are common to all cells; R and C are not models without any prescribed mechanics.

The generic residual acts on all four canonical coordinates $x = (q, p)$. Its inputs use componentwise RMS scales computed from the fourteen training trajectories, and its energy scale is the training RMS of H_2. A common evenization and subtraction at the origin impose $N_{\theta,s}(-x) = N_{\theta,s}(x)$ and $N_{\theta,s}(0) = 0$. No degree-six prefactor or independent quadratic skip is present. In particular, the

residual Hessian is free, so the no-prior cells are not structurally prevented from approximating the missing quadratic term. This is an approximation task for finite-width networks, not a claim of exact quadratic representation at arbitrary parameter values.

The real network has two width-24 hidden layers with real tanh and a linear readout, giving 744 trainable real coordinates. The complex network has two width-16 hidden layers with literal complex tanh, independent $z, \bar{z}$ affine inputs, and a real readout, giving 736 real coordinates. Its layerwise imaginary-argument budgets are 0.15 and 0.6. These constraints apply on the declared normalized phase box containing all training states. The counts differ by approximately 1.09%; there is no parameter padding. Widths, parameter domains and initialization laws also differ. Within a representation and seed, the prior and no-prior cells start with identical raw network parameters; real and complex initial functions are not matched.

All cells use the same frozen 42-coordinate complex PSD matrix head and the same distributed-viscosity cohort of fourteen trajectories on $[0, 0.4]$, with 801 observation nodes per trajectory. The earlier configuration-only structured model used a different conservative $[0, 0.2]$ cohort; it is a separate baseline. The matrix head is the previously identified distributed-viscosity head, fitted with that structured Hamiltonian frozen; it is not refitted for these comparators. Only time, configuration, observed velocity and prescribed input enter the fit. Observed momentum is reconstructed as $G(q)v$. The predicted dynamics use

$$\begin{aligned} q' &= H_{s,p}, \\ p' &= -H_{s,q} - \widehat{R}(q)H_{s,p} + ug(q). \end{aligned} \tag{26}$$

Consequently, dissipation uses the model's actual $H_{s,p}$, and the kinematic discrepancy is $H_{s,p} - v$; observed velocity is not silently substituted into the predicted damping force. Twenty windows per trajectory use the same actual-time, five-point Newton–Cotes quadrature. The common scalar loss is the mean normalized momentum-weak square plus 0.1 times the quadrature-weighted normalized kinematic square and a raw-coordinate penalty with coefficient $10^{-20}/2$. All scales come from the fourteen training trajectories. No separate potential-nuisance fit, reference-force labels or validation selection is used.

Seeds 43, 47 and 53 give twelve fits. Each completes 400 full-batch Adam updates with learning rate 0.01, $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon = 10^{-8}$. The lowest finite training total among the initial state and the 400 recorded updates selects each checkpoint. All twelve selected outputs are frozen before this factorial evaluation, and their training-loss replays agree. There are no failed fits, no failed forty-point model evaluations and no failed model–holdout evaluations among the 48 latter combinations. The three seeds are optimization repeats on common data, rather than independent physical datasets. The initial functions also differ materially between representations: for seed 43, the initial training totals for R, C, GR and GC are 1.197470, 1.085593, 0.139463 and 0.001348, respectively. The common learning rate and update budget therefore do not remove the effects of the different initialization and constraint-induced function scales, especially in the prior-assisted pair.

The primary evaluation uses the same forty sealed observation-only gradient labels Y_i at the prescribed rest points $(Q_i, 0)$:

$$E_s^2 = \frac{1}{40}\sum_{i=1}^{40} \left\| H_{s,q}(Q_i, 0) - Y_i \right\|_2^2. \tag{27}$$

The forty labels and four original holdout trajectories had already served other analyses. They are therefore reused validation, not a new blind confirmation of these generic comparators. At $p = 0$, the metric tests potential gradients; it does not validate arbitrary phase-space derivatives. Table 1 retains every seed.

The prior improves both representations in every paired seed, and the complex fit has lower RMSE within both prior settings. The interaction is negative for every seed: the prior produces the larger absolute error reduction for the real network and narrows the absolute real–complex gap. These observations describe this data, architecture pair and optimization budget. They do not isolate complex arithmetic from the different widths, constraints, initialization and parameter-dependent regularization, or establish universal complex superiority.

Full-phase consistency is assessed separately on the four reused holdout trajectories, using the fixed training scales, the same frozen matrix head and no additional coefficients. Table 3 pools squared normalized errors equally across the three seeds and four trajectories before taking the square root. It reports weak momentum and quadrature-weighted $H_{s,p} - v$ discrepancies, not a new integrated rollout. The last column gives the range across seeds of $\{40^{-1}\sum_i \|H_{s,p}(Q_i, 0)\|_2^2\}^{1/2}$. Although the reference velocity vanishes at these rest points, the generic residual can contain mixed q, p terms. Central evenization does not impose evenness in p at fixed q, and the nonzero rest velocity

Table 1. Generic-network gradient-vector RMSE on the same forty reused rest-point labels. Entries are $10^3 E_s$, in common dimensionless potential-gradient units. The final row is the arithmetic seed mean.

Seed	R	C	GR	GC
43	7.219805	6.034963	0.099531	0.009317
47	9.520357	6.648840	0.298038	0.007967
53	12.884718	5.769706	1.104723	0.010986
Mean	9.874960	6.151169	0.500764	0.009423

Table 2. Five paired contrasts of the RMSE in Table 1, multiplied by 10^3. For $A - B$, a positive value means that B has smaller error. The interaction is $(\mathrm{GR} - \mathrm{GC}) - (\mathrm{R} - \mathrm{C})$. Means use all three paired seeds.

Contrast	Seed 43	Seed 47	Seed 53	Mean
R−GR	7.120274	9.222319	11.779995	9.374196
C−GC	6.025646	6.640873	5.758720	6.141746
R−C	1.184843	2.871517	7.115013	3.723791
GR−GC	0.090214	0.290071	1.093737	0.491341
Interaction	−1.094628	−2.581446	−6.021276	−3.232450

diagnostic remains relevant even when the potential-gradient error is relatively small.

The absolute-accuracy limitation is substantial. On the identical forty labels, the zero-network fourth-order baseline $H_4 = T_{\mathrm{nl}} + H_2 + V_{\mathrm{quartic}}$ has gradient-vector RMSE approximately 1.297917×10^{-11}. Every generic GC fit is worse: its RMSE ranges from 7.966939×10^{-6} to 1.098578×10^{-5}, approximately 6.1×10^5 to 8.5×10^5 times the baseline error. The generic prior-assisted family contains H_4 at zero residual, so this gap is a limitation of the stated generic training protocol at the resolved signal scale, not proof that the physical prior is unrepresentable. Relative improvement within the factorial experiment therefore does not demonstrate improvement over the prescribed fourth-order physics. The precision-centered comparison below tests whether output scaling and centering reduce this absolute-accuracy gap with the full-phase network.

These comparisons use ordinary floating-point network evaluation. Exact rational propagation of the sealed label widths supplies sensitivity bounds conditional on the returned network numbers; it does not enclose neural evaluation and differentiation errors and is not a strict network-ranking certificate. These twelve factorial outputs are distinct from the selected precision-centered GC output certified in Section 4. Their relative ranking does not transfer any function-specific certificate to them.

3.3 A precision-centered prior-assisted comparison

A separate two-fit comparison tests whether objective and output scaling account for part of the absolute-accuracy gap in Section 3.2. It retains the generic full-phase real and complex architectures, with 744 and 736 trainable real coordinates, respectively, and uses seed 43 only. Both models now have the actual scalar form $H_4 + N_s$, with the same finite-Gauss kinetic energy and fourth-order potential. An exactly zero readout makes their initial functions identical to H_4; there is no degree-six multiplier, polynomial replacement or separately fitted potential block.

The output multipliers are $\rho_{\mathrm{GR}} = 2^{-32}$ and $\rho_{\mathrm{GC}} = 2^{-28}$. They are dyadic scales derived from the training residual and readout-Jacobian singular values, using only the same fourteen distributed-viscosity trajectories. The momentum-weak balance is centered about the actual scalar H_4 before adding the network perturbation. The perturbation is evaluated directly, rather than obtained by subtracting full and baseline floating-point forces. The baseline still contains the arithmetic of the native observations and quadrature. Its quadratic jet is evaluated from the actual scalar H_4, rather than regrouped through the stored approximation to K_2. The sum of the normalized weak loss and the 0.1-weighted kinematic loss is divided by its fixed H_4 value. There is no raw-parameter penalty. The matrix head remains fixed, and the kinematic and damping terms retain the actual full-phase $H_{s,p}$.

Both fits complete 400 full-batch Adam updates with learning rate 0.01 and $\epsilon = 10^{-8}$. Every layer remains trainable. The initial zero readout gives zero hidden-layer data gradients at the first update; nonzero hidden-parameter and prediction changes are recorded after the second update. Checkpoint selection again minimizes the training total over the initial state and all recorded updates. The selected updates are 400 for GR and 385 for GC, with training totals relative to the initial value of approximately 4.1951×10^{-4} and 1.9017×10^{-2}, respectively.

Both selected outputs are frozen before their evaluation against the same forty previously used labels. No additional held-out dataset is introduced or used for selection in this

Table 3. Reused full-phase diagnostics. Weak and kinematic columns are pooled normalized RMS values over twelve seed–trajectory combinations per cell. The rest-velocity range covers the three seedwise forty-point RMS values and is in the original dimensionless velocity coordinates.

Cell	Momentum weak	$H_{s,p} - v$	Rest $H_{s,p}(Q,0)$ RMS range
R	5.310319×10^{-2}	1.255422×10^{-1}	$[1.518317 \times 10^{-3},\ 5.222680 \times 10^{-3}]$
C	3.948955×10^{-2}	6.424029×10^{-2}	$[1.484900 \times 10^{-3},\ 2.417393 \times 10^{-3}]$
GR	4.187033×10^{-3}	1.810586×10^{-2}	$[1.498658 \times 10^{-4},\ 3.086894 \times 10^{-3}]$
GC	5.695685×10^{-5}	3.301787×10^{-4}	$[8.301210 \times 10^{-6},\ 2.276685 \times 10^{-5}]$

comparison. It is exploratory reused validation, rather than a new blind test, and does not replace the twelve fits or five paired contrasts above. Only these forty labels are evaluated; no trajectory-holdout evaluation is performed for the two repaired outputs.

The real model has the smaller error in this comparison, and neither learned output improves on the shared prior (Table 4). Together with the original factorial result, this reversal demonstrates sensitivity to the stated training scales and initialization; it does not establish an intrinsic advantage of either representation. The selected precision-centered GC output is the frozen function in (7). Its subsequent observation-error, conservative-torus and physical-feedback certificates are proved for this checkpoint in Sections 3.4, 4 and 5. They do not use a certificate belonging to the earlier structured model.

3.4 A deterministic observation-to-function error bound

An empirical weak loss controls observations along the available curves. To obtain an error on a complex state domain, we additionally certify an observation inverse and the actual remainder of the selected network. Throughout this subsection the output is the same 736-coordinate complex model used in the deployed energy. Set

$$e = N_\vartheta - U,$$
$$\Omega = \left\{ x \in \mathbb{C}^4 : \begin{matrix} |\operatorname{Re} x_j| \le 3/50, \\ |\operatorname{Im} x_j| \le 1/100 \end{matrix} \right\}.$$

The functions have analytic extensions to a neighbourhood of this box.

Let χ contain every nonconstant monomial of even total degree at most six in the normalized coordinates x_j/s_j. There are 129 columns: 15 configuration-only columns and 114 containing a momentum factor. Choose an explicitly specified polynomial $P_N = \chi^{\mathsf{T}} a_N$ and write

$$\begin{aligned} U &= U_6 + \rho_U, \\ P_N &= N_\vartheta - \rho_N, \\ U_6 &= \frac{(d^{\mathsf{T}} q)^6}{720}. \end{aligned} \tag{28}$$

The polynomial is an auxiliary representative of the fitted function; the trained network and its hidden layers remain unchanged.

The observation operator $\mathcal{L}$ uses configuration derivatives at rest anchors $(Q_i, 0)$ and momentum derivatives at the represented chart points $(\bar{q}_l, G(\bar{q}_l)\bar{v}_l)$. At rest, sealed labels satisfy $Y_i = \nabla(V_4 + U)(Q_i) + \xi_i$. The signed residuals are

$$\begin{aligned} r_{q,i} &= N_{\vartheta,q}(Q_i, 0) \\ &\quad + V_{4,q}(Q_i) - Y_i, \\ r_{p,l} &= N_{\vartheta,p}(\bar{q}_l, G(\bar{q}_l)\bar{v}_l). \end{aligned} \tag{29}$$

Only the selected components are retained. Since $U_p = 0$, the second reference label is structural information from the shared kinetic graph; it does not require exported native momentum or assert that the chart point equals an exact physical trajectory state. With zero moving entries in ξ, the exact identity is $r = \mathcal{L}e - \xi$.

The selected physical derivative rows form a square design $A = \mathcal{L}\chi^{\mathsf{T}}$. After a common column permutation, $A = A_q \oplus A_p$ with sizes 15 and 114. Configuration derivatives at $p = 0$ annihilate mixed monomials, while momentum derivatives annihilate configuration-only monomials. Interval evaluation encloses the exact observed chart, and a fixed exact dyadic matrix R satisfies

$$\begin{aligned} \sup_A \|I - RA\|_\infty &\le \eta_A < 1, \\ \mathcal{J}_A &= (RA)^{-1} R, \\ \kappa_A &= \frac{\eta_A}{1 - \eta_A}. \end{aligned} \tag{30}$$

In particular, $\mathcal{J}_A A = I$. Physical derivative rows retain their $1/s_j$ factors. If row weights are introduced, the same weights must multiply residuals, label errors and point remainders.

Table 4. Precision-centered comparison on the same forty reused rest labels. Gradient-vector RMSE uses the normalization in (27). Neural derivatives are ordinary floating-point scalar-energy automatic derivatives; the table is not an interval ranking certificate.

Model	Selected update	Rest-gradient vector RMSE
GR	400	$1.427681587 \times 10^{-11}$
GC	385	$9.819100531 \times 10^{-11}$
Shared H_4	—	$1.297916885 \times 10^{-11}$

Put $t_N = \mathcal{L}\rho_N$, $t_U = \mathcal{L}\rho_U$ and $F_\alpha = \partial_x^\alpha \chi^{\mathsf{T}}$. Applying $\mathcal{L}$ to (28) and using the residual sign gives

$$\begin{aligned} A(a_N - a_U) &= r + \xi - t_N + t_U, \\ \partial_x^\alpha e &= F_\alpha \mathcal{J}_A (r - t_N + t_U) \\ &\quad + F_\alpha \mathcal{J}_A \xi \\ &\quad + \partial_x^\alpha \rho_N - \partial_x^\alpha \rho_U. \end{aligned} \tag{31}$$

Here a_U is the coefficient vector of U_6. It is used only in verification. Both observation-point remainders and target-domain remainders occur in the identity.

For $|x_j| \le r_j$, define the physical derivative envelopes

$$\begin{aligned} g_{\alpha\beta} &= \prod_{j=1}^{4} \frac{(\beta_j)_{\alpha_j} r_j^{\beta_j - \alpha_j}}{s_j^{\beta_j}}, \\ \mathcal{G}_\alpha(v) &= \sum_\beta g_{\alpha\beta} v_\beta \\ &\quad + \kappa_A \left(\sum_\beta g_{\alpha\beta} \right) \|v\|_\infty, \end{aligned}$$

where a term is zero when $\beta_j < \alpha_j$. On Ω one may take $r_j = \sqrt{37}/100$. The Neumann decomposition $\mathcal{J}_A y = Ry + [(RA)^{-1} - I]Ry$ then yields the computable bound

$$\begin{aligned} \|\partial_x^\alpha e\|_\Omega &\le \mathcal{G}_\alpha\big(|R(r - t_N + t_U)|\big) \\ &\quad + \mathcal{G}_\alpha(|R|\zeta) \\ &\quad + \|\partial_x^\alpha \rho_N\|_\Omega + \|\partial_x^\alpha \rho_U\|_\Omega, \\ &\quad |\xi| \le \zeta. \end{aligned} \tag{32}$$

The signed sum is enclosed before absolute values are taken. Replacing $\mathcal{J}_A$ by R without its Neumann correction, or discarding a point remainder, would invalidate this bound. The full signed variant and input-uncertainty conditions are given in Supplementary Lemma S9.

For the selected output, the 129-row certificate supplies all 34 multi-index bounds $b_\alpha \ge \|\partial_x^\alpha e\|_\Omega$ of orders one through three. For $Y = -J_4 \nabla e$, define

$$\begin{aligned} \delta_0 &= \max_i b_{\mathbf{e}_i}, \\ \delta_1 &= \max_i \sum_j b_{\mathbf{e}_i + \mathbf{e}_j}, \\ \delta_{t1} &= \sum_{i,j} b_{\mathbf{e}_i + \mathbf{e}_j}, \\ \delta_2 &= \max_i \sum_{j,k} b_{\mathbf{e}_i + \mathbf{e}_j + \mathbf{e}_k}. \end{aligned} \tag{33}$$

These bound the field error, its derivative, the transposed derivative and its ordered bilinear second derivative. In particular, all physical momentum derivatives are retained. The certified outward bounds are

Quantity	Upper bound on Ω
δ_0	1.898×10^{-8}
δ_1	2.368×10^{-6}
δ_{t1}	6.398×10^{-6}
δ_2	5.777×10^{-4}

The derivative sums include all ordered multiplicities. These are deterministic bounds for the specified numerical reference and realized observations. The reference fidelity allowance is verified after label sealing and is not supplied to the optimizer. Consequently dependent training curves do not obstruct this pathwise conclusion. Identification of an uncertain measured plant requires additional fidelity budgets.

3.5 A prospective sampling statement and its scope

The fourteen trajectories are dependent time records. They are not assigned an iid phase-space law. A separate prospective construction fixes the normalized coordinates and sampling densities before collection. For the two column blocks, write

$$\begin{aligned} B_b &= D_b \nabla_{x,b} \chi_b^{\mathsf{T}}, \\ M_b &= \mathbb{E}_{\mathrm{unif}} B_b^{\mathsf{T}} B_b, \\ K_b &= \mathrm{tr}(M_b^{-1} B_b^{\mathsf{T}} B_b), \\ b &= q, p, \end{aligned}$$

where $D_q = \mathrm{diag}(s_1, s_2)$ and $D_p = \mathrm{diag}(s_3, s_4)$. The excluded derivative nullspaces imply $M_b \succ 0$. Under independent anchors with density π_b, the weighted rows are

$\pi_b^{-1/2} B_b$; residual, fidelity and remainder rows receive the identical $\pi_b^{-1/2} D_b$ transformation. If $K_b/\pi_b \le L_b$, then

$$\begin{gathered} n_b \ge \frac{2L_b}{\varepsilon_b^2} \log \frac{d_b}{\alpha_b} \\ \Longrightarrow \\ \frac{A_b^{\mathsf{T}} A_b}{n_b} \succeq (1-\varepsilon_b) M_b \end{gathered} \tag{34}$$

on a common event of probability at least $1 - \alpha_q - \alpha_p$. Here $d_q = 15$, $d_p = 114$ and n_b counts independent anchors, each contributing one two-row positive-semidefinite summand. The full-support choice $\pi_b = (1 + K_b/d_b)/2$ permits $L_b = 2d_b$.

The matrix event controls every coefficient vector. Together with a common fidelity event and each output's sound remainder certificate, it therefore controls outputs chosen using the same data. If acceptance also verifies the conservative-family conditions below, the precise probability statement is

$$\begin{gathered} \mathbb{P}\left\{ \begin{array}{c} \text{an accepted output has a false} \\ \text{certified conclusion} \end{array} \right\} \\ \le \alpha_{\mathrm{fid}} + \alpha_q + \alpha_p. \end{gathered} \tag{35}$$

Supplementary Theorem S11 gives the derivative transport and proof. This prospective result has not been instantiated by the existing trajectories. It provides neither conditional confidence given acceptance nor a guarantee that an optimizer finds an acceptable output; the deterministic bounds above do not require either assertion.

4 Conservative certification and continuation to the reference

4.1 An invariant torus of the learned full-phase energy

The conservative statement concerns the deployed scalar $\widehat{H}_\vartheta = T_G + V_4 + N_\vartheta(q,p)$, with the stored parameters interpreted as exact real constants in the constrained network graph. The complexification treats q_1, q_2, p_1, p_2 as independent variables. Dissipation and the magnetic input are excluded from this Hamiltonian vector field.

For period-one angles fix

$$\begin{gathered} \nu = \frac{\omega_1}{2\pi}(1,\alpha), \\ \omega_1 = \frac{499974648628101}{500000000000000}, \\ \alpha = \frac{2524851\phi_g + 622102}{410497\phi_g + 101143}, \\ \phi_g = \frac{1+\sqrt{5}}{2}. \end{gathered} \tag{36}$$

The bounded continued fraction with its golden tail gives an all-integer Diophantine certificate

$$\begin{gathered} |k\cdot\nu| \ge \gamma |k|_1^{-1}, \\ \gamma = \frac{4828522443384859}{5764607523034234 88}, \\ k \in \mathbb{Z}^2 \setminus \{0\}. \end{gathered} \tag{37}$$

The corresponding period-one cohomological estimate uses $\tau = 1$ and $c_R = 1/7$. For matrices, the analytic norm is $\|A\|_\rho = \max_i \sum_j \sup_{\mathbb{T}^2_\rho} |A_{ij}|$; transpose bounds are supplied separately.

Theorem 2. The specified learned conservative Hamiltonian. *For the selected 736-coordinate complex model and the prescribed finite Gauss mechanical graph, there exists a real analytic invariant two-torus with frequency* (36)*. The certified 8064-mode rational Fourier polynomial K_0 has a corrected embedding K_∞, holomorphic on $\mathbb{T}^2_{1/400}$, such that*

$$\begin{gathered} J_4 \nabla \widehat{H}_\vartheta(K_\infty) = DK_\infty \nu, \\ \|K_\infty - K_0\|_{1/400} < 9.018 \times 10^{-24}. \end{gathered} \tag{38}$$

Its real restriction is an embedding, and the complex image lies in $|\operatorname{Re} x_j| < 3/50$, $|\operatorname{Im} x_j| < 1/100$ *for all four canonical coordinates.*

Proof. The exact-parameter network evaluation and finite Gauss graph give a real-type holomorphic Hamiltonian on the common domain, with the required field derivatives. Set

$$\begin{gathered} L = DK_0, \\ B = (L^{\mathsf{T}} L)^{-1}, \\ \mathcal{N} = J_4 L B, \\ T = \mathcal{N}^{\mathsf{T}} J_4 \{ DX_{\widehat{H}_\vartheta}(K_0)\mathcal{N} - D_\nu \mathcal{N} \}. \end{gathered}$$

Whole-strip Fourier bounds verify the image, Gram inverse and rank. A complete continuous-mean calculation gives $\|\langle T\rangle^{-1}\|_\infty < 19026.13$; it includes the full 4×4 Hessian and the directional derivative of the normal frame. Analytic transposes, rather than conjugate transposes, enter the frame.

The residual $E_0 = X_{\widehat{H}_\vartheta}(K_0) - DK_0\nu$ is enclosed on $\rho = 1/40$. An outer-domain certificate on $\widehat{\rho} = 1/10$, followed by a 448×448 outward-arithmetic Fourier transform, retains sample radii, every representative mode and the infinite angle aliases. It yields

$$\|E_0\|_{1/40} < 1.196 \times 10^{-42}.$$

The canonical $n = d = 2$ Case III theorem of Haro and Luque (2019, Theorem 3.1) then applies, with the arXiv version 1 conventions. Each auxiliary geometric bound is chosen as $11/10$ of its certified input bound. Recomputing all constants gives a sufficient residual threshold greater than 1.722×10^{-39} and a maximum of all eight strict gates below 6.942×10^{-4}. The resulting displacement is bounded as in (38), remains within the domain and preserves the real embedding by the corrected harmonic phase test.

For clarity, the theorem's symplectic matrix is $\Omega_{\mathrm{HL}} = -J_4$ and its compatible normal is $-\mathcal{N}$. Its torsion is therefore $-T$; the mean inverse norm is unchanged. The theorem's auxiliary Riemannian metric is I_4, distinct from the mechanical inertia $G(q)$ inside the Hamiltonian. These identifications supply the claimed canonical specialization without assuming exact Lagrangianity of K_0.

This conclusion belongs to the fixed learned conservative energy. The observation error bounds and a continuation argument are additionally needed to establish a torus of the reference H_*.

4.2 Observation-to-reference continuation

Define the exact family

$$\begin{aligned} H_\lambda &= \widehat{H}_\vartheta + \lambda(U - N_\vartheta), \\ X_{H_\lambda} &= X_{\widehat{H}_\vartheta} + \lambda Y, \\ 0 &\le \lambda \le 1, \end{aligned} \tag{39}$$

with $Y = -J_4\nabla e$ as in Section 3.4. If $(\widehat{c}_0, \widehat{c}_1, \widehat{c}_{t1}, \widehat{c}_2)$ bound the actual learned field and its derivatives on Ω, then

$$\begin{aligned} &(c_0, c_1, c_{t1}, c_2) = \\ &\quad (\widehat{c}_0 + \delta_0, \widehat{c}_1 + \delta_1, \\ &\qquad \widehat{c}_{t1} + \delta_{t1}, \widehat{c}_2 + \delta_2) \end{aligned} \tag{40}$$

are uniform real-parameter inputs. This is the direct use of the observation-based learning error in the conservative theorem.

An explicit approximate embedding must also follow the family:

$$\begin{aligned} P_m(\varphi, \lambda) &= K_0(\varphi) + \sum_{j=1}^{m} \lambda^j U_j(\varphi), \\ \mathcal{R}_m &= X_{H_\lambda}(P_m) - D_\nu P_m. \end{aligned} \tag{41}$$

Every U_j and all later certificates refer to the complete preceding prefix. The field increments in (40) do not replace a bound on this full invariance defect.

Suppose $\mathcal{R}_m$ is holomorphic on $\mathbb{T}^2_{\widehat{\rho}} \times \{|\lambda| \le R\}$, $\widehat{\rho} > \rho$, $R > 1$, with modulus at most M. For certified coefficients $\epsilon_j \ge \|[\lambda^j]\mathcal{R}_m\|_\rho$ at indices $S \subseteq \{0, \dots, L\}$, a complete bound is

$$\begin{aligned} E = &\sum_{j \in S} \epsilon_j \\ &+ \sum_{\substack{0 \le j \le L \\ j \notin S}} \frac{M}{R^j} \\ &+ \frac{M}{R^L(R-1)}. \end{aligned} \tag{42}$$

This follows from Cauchy's estimate and the infinite geometric tail. Nonlinear composition generates arbitrarily high parameter coefficients even when P_m is finite. Each computed coefficient must also include its complete angle interpolation and alias errors; uncomputed low coefficients cannot be set to zero.

For each real parameter, put $L_\lambda = D_\varphi P_m$, $B_\lambda = (L_\lambda^{\mathsf{T}} L_\lambda)^{-1}$ and $\mathcal{N}_\lambda = J_4 L_\lambda B_\lambda$. The required mean is that of this varying embedding:

$$\begin{aligned} T_\lambda &= \mathcal{N}_\lambda^{\mathsf{T}} J_4 \\ &\quad \cdot \{DX_{H_\lambda}(P_m)\mathcal{N}_\lambda - D_\nu \mathcal{N}_\lambda\}, \\ I &\ge \sup_{\lambda \in [0,1]} \|\langle T_\lambda \rangle^{-1}\|_\infty. \end{aligned} \tag{43}$$

The fixed-K_0 inverse does not by itself establish this condition. Signed integration with complete errors, or a certified transport to the same P_m, is necessary.

To state the remaining quantitative requirements, let k, k^t, b bound $L_\lambda, L_\lambda^{\mathsf{T}}, B_\lambda$ on the whole strip, and let $r_{\mathrm{dom}} > 0$ be a common tube radius in Ω. Choose $\sigma_K > k$, $\sigma_{K^t} > k^t$, $\sigma_B > b$, $\sigma_T > I$. With admissible iteration parameters $a_1 > 1$, $\nu_{\mathrm{it}} > 0$ and $\ell = \rho/a_3$, recompute the Case III constants from these actual uniform inputs. Writing $C_K = C_{\Delta K}$, the

eight strict tests can be expressed compactly as

$$\begin{aligned}
&\frac{(a_1a_3)^{4\tau}C_E E}{\gamma^4\rho^{4\tau}} < 1,\\
&\frac{E}{\ell\nu_{\mathrm{it}}} < 1,\\
&\frac{2C_{\mathrm{sym}}E}{\gamma\ell^{\tau+1}} < 1,\\
&\frac{E}{\gamma^2\ell^{2\tau+1}(1-a_1^{1-2\tau})}\\
&\times \max\left\{\begin{array}{c}\dfrac{2C_K}{\sigma_K-k},\\ \dfrac{4C_K}{\sigma_{K^t}-k^t},\\ \dfrac{C_B}{\sigma_B-b},\\ \dfrac{C_{T^{-1}}}{\sigma_T-I}\end{array}\right\} < 1,\\
&\frac{C_K E}{r_{\mathrm{dom}}\gamma^2\ell^{2\tau}(1-a_1^{-2\tau})} < 1.
\end{aligned} \tag{44}$$

The four terms inside the maximum are four preservation tests. The remaining tests control contraction, iteration, approximate symplectic invertibility and the domain. The constants, strip-loss relation and corrected embedding criterion are specified in Supplementary Section S5.

Theorem 3. Conditional conservative reference acceptance. *Assume that the observation error and input fidelity bounds hold for the selected output. Suppose the same complete P_m has a uniform analytic tube, whole-strip Gram inverse, actual mean inverse (43), and complete defect bound (42). If the freshly computed eight tests (44) and the corrected real embedding test pass, then for every real $\lambda \in [0,1]$ there is an invariant real analytic embedding K_λ^* with frequency ν and*

$$\begin{aligned}
&\|K_\lambda^* - P_m(\cdot,\lambda)\|_{\rho_\infty}\\
&\quad\le \frac{C_2 E}{\gamma^2\rho^{2\tau}}.
\end{aligned}$$

In particular K_1^* is a torus of the specified conservative finite Gauss reference H_*.

Proof. The observation identity and (32) provide the full physical derivatives of e, hence the uniform field inputs (40). For each fixed real λ, the common domain, Diophantine bound, Gram and actual torsion conditions are the hypotheses of the canonical a posteriori theorem. The eight tests preserve those bounds during its iteration. Its correction is uniform because all inputs are uniform. The additional real embedding test includes both the C^0 correction and its retained-strip Cauchy derivative allowance. Finally $H_1 = T_G + V_4 + U = H_*$ exactly, including cancellation of the entire learned momentum derivative. This proves the endpoint assertion. Supplementary Theorem S11 also gives the common-event version for data-dependent selection.

4.3 Verified continuation for the deployed network

For the selected GC output, the complete degree-ten prefix P_{10} satisfies the hypotheses of Theorem 3. The observation bounds following (33) enter the uniform field constants in (40). The varying geometry and the continuous signed mean are certified for this same P_{10}, with $I < 25539.514$. The full defect, including ordinary parameter coefficients through order fourteen and the infinite tail, satisfies

$$\begin{aligned}
E &< 3.848\times10^{-41},\\
E_{\mathrm{suff}} &> 9.303\times10^{-40},\\
\max_{1\le j\le 8}\mathrm{gate}_j &< 0.04136 < 1.
\end{aligned} \tag{45}$$

Here E_{suff} is a sufficient upper limit on an admissible defect; it also includes the domain and corrected-embedding budgets.

Theorem 4. The specified conservative reference family. *For every real $\lambda \in [0,1]$, the Hamiltonian (39) has a real analytic embedded invariant two-torus with the fixed frequency (36). Its embedding is holomorphic on $\mathbb{T}^2_{1/400}$ and satisfies*

$$\begin{aligned}
X_{H_\lambda}(K_\lambda^*) &= DK_\lambda^*\nu,\\
\|K_\lambda^* - P_{10}(\cdot,\lambda)\|_{1/400} &< 3.949\times10^{-22}.
\end{aligned} \tag{46}$$

In particular, the specified finite-Gauss conservative mechanical reference H_* has such a torus at $\lambda = 1$.

Proof. The certified full-phase observation inverse supplies the field increments. The actual prefix has a common analytic domain, a whole-strip Gram inverse, and the continuous mean inverse stated above. Coefficients zero and one of its defect use their completed whole-strip certificates. Each of the thirteen higher coefficients uses a complete outward-arithmetic Fourier enclosure, retaining sample radii, Nyquist representatives and all infinite angular aliases. A direct product-domain bound at parameter radius 512 controls every parameter coefficient above order fourteen. Summing the exact rational endpoints gives (45).

Independent recomputation verifies all eight strict inequalities and the domain correction. The corrected real-embedding phase bound is less than $0.005686 < 1/2$. Applying Theorem 3 separately at each real parameter gives (46). The exact cancellation $H_1 = T_G + V_4 + U = H_*$ identifies

the reference endpoint. Supplementary Section S7 gives the coefficient bounds and numerical application.

The quantifier is existence for each real parameter with common frequency and error bounds. The theorem does not assert analytic parameter dependence of the exact tori. Dissipation and magnetic feedback are treated through the separate power and safety arguments in Section 5.

5 Dissipative structure and sampled magnetic safety

The conservative certificate in Section 4 and the feedback analysis use the same frozen full-phase energy $\widehat{H}$. A separately fitted, configuration-only complex head supplies the dissipation. The resulting composition comprises the 736-parameter energy network and the 42-parameter damping head; it is not a joint fit of the two networks. All derivatives below use the canonical state $x = (q, p)$ and the dimensionless time τ.

5.1 Learned and reference power

On the real state domain let $G_8 = SS^{\mathsf{T}} \succ 0$ and define

$$\widehat{M}_\psi(q) = \mathrm{Re}\{A_\psi(q)A_\psi(q)^*\},$$
$$\widehat{R}(q) = S(q)\widehat{M}_\psi(q)S(q)^{\mathsf{T}}.$$

For real z, $z^{\mathsf{T}}\widehat{M}_\psi z = \|A_\psi^* z\|_2^2 \geq 0$; hence the damping matrix is real symmetric positive semidefinite. With $\widehat{v} = \widehat{H}_p = G_8^{-1}p + N_p$ as in (8), the learned equations are

$$q' = \widehat{v},$$
$$p' = -\widehat{H}_q - \widehat{R}\widehat{v} + g(q)u,$$
$$g(q) = d\cos(d^{\mathsf{T}}q).$$

Cancellation of the Hamiltonian terms gives

$$\widehat{H}' = -\widehat{d} + u\widehat{y}, \qquad \widehat{d} = \widehat{v}^{\mathsf{T}}\widehat{R}\widehat{v} \geq 0, \quad \widehat{y} = g^{\mathsf{T}}\widehat{v}. \tag{47}$$

This identity concerns the learned dynamics. The reference plant in (6) instead satisfies

$$q' = v_* = G_8^{-1}p, \quad p' = -H_{*,q} - R_*v_* + g(q)u,$$
$$H_*' = -d_* + uy_*, \quad d_* = v_*^{\mathsf{T}}R_*v_*, \quad y_* = g^{\mathsf{T}}v_*. \tag{48}$$

Here $R_* = SM_*S^{\mathsf{T}} \succeq 0$ is the declared distributed linear-viscosity law. Put $e = \widehat{H} - H_* = N - U$. Then $e_p = \widehat{v} - v_* = N_p$ is generally nonzero. The reference-power balance, including this velocity difference, is the basis of the safety filter.

Suppose same-function certificates give $|e| \leq \varepsilon_H$, $|e_{p,j}| \leq \varepsilon_{p,j}$ and $\|\widehat{M}_\psi - M_*\|_2 \leq \delta_M$ on the declared real domain. For $z = S^{\mathsf{T}}\widehat{v}$ and $\eta = S^{\mathsf{T}}e_p$, take bounds $a \geq \|z\|_2$, $b \geq \|\eta\|_2$ and $\bar{m} \geq \|\widehat{M}_\psi\|_2$. Expanding $(z-\eta)^{\mathsf{T}}M_*(z-\eta)$ and using $\widehat{M}_\psi \succeq 0$ yields

$$|y_* - \widehat{y}| \leq \sum_j |d_j|\varepsilon_{p,j},$$
$$d_* \geq \widehat{d} - 2\bar{m}ab - \delta_M(a+b)^2. \tag{49}$$

One may use $a^2 \geq \widehat{v}^{\mathsf{T}}G_8\widehat{v}$ and $b^2 \geq \varepsilon_p^{\mathsf{T}}|G_8|\varepsilon_p$. The first reserve transports the magnetic port; both reserves in the second line are retained in the dissipation credit.

The absence of e_q from (48) does not set that error to zero. In fact, along the reference plant the learned storage obeys

$$\widehat{H}' = -\widehat{d} + u\widehat{y} + e_q^{\mathsf{T}}\widehat{v} + e_p^{\mathsf{T}}(\widehat{R}\widehat{v} - \widehat{H}_q) + \widehat{v}^{\mathsf{T}}(\widehat{R} - R_*)v_*.$$

Thus its zero-input derivative cannot be inferred from (47) along the reference plant. The physical-energy ledger below uses (48) and the complete transport bounds (49).

5.2 Observation enclosures and the held-input filter

At sample τ_k, the filter requires an enclosing canonical state box C_k containing the true reference state. If position and velocity are observed instead, a sound transformation through $p = G_8(q)v_*$ must supply this box. Interval evaluation of the frozen learned functions gives a port estimate $\widetilde{y}_k$, an error radius $\epsilon_{y,k}$ and a physical dissipation credit $\underline{d}_k$. They are chosen so that

$$\epsilon_{y,k} \geq \sup_{C_k} |\widehat{y} - \widetilde{y}_k| + \sum_j |d_j|\varepsilon_{p,j},$$
$$\underline{d}_k = \max\{0, \widehat{d}_k^- - 2\bar{m}_k\bar{a}_k\bar{b}_k - \delta_M(\bar{a}_k + \bar{b}_k)^2\},$$

with the first quantity rounded upward and the second downward. Here $\widehat{d}_k^-$ is a lower bound over C_k, and all other quantities bound their respective values throughout that box. Consequently $|y_*(\tau_k) - \widetilde{y}_k| \leq \epsilon_{y,k}$ and $0 \leq \underline{d}_k \leq d_*(\tau_k)$.

Let $c < E_g$, and let the designated component of $\{H_* \leq E_g\}$ be compact and strictly contained in the state, observation and clearance domain. For every admissible constant command suppose the reference derivatives satisfy $|y_*'| \leq L_y$ and $|d_*'| \leq L_d$ before first guard exit. Initialize a ledger with

$$H_*(x_0) \leq W_0 = \left\lceil \max\{0, \widehat{H}^+(x_0) + \varepsilon_H\} \right\rceil_{\Delta_W} < c.$$

For an uncertain initial state the upper evaluation must cover its entire initial enclosure. For an exact hold length h_k and $0 < \alpha < 1$, admit an actual represented nonzero command only when $|u_k| \le u_{\max}$ and

$$I_k(u_k) = h_k(u_k\widetilde{y}_k + |u_k|\epsilon_{y,k} - \underline{d}_k) + \tfrac{1}{2}h_k^2(|u_k|L_y + L_d) \le \alpha(c - W_k). \tag{50}$$

Set $W_{k+1} = \lceil \max\{0, W_k + I_k(u_k)\} \rceil_{\Delta_W}$ and independently require $W_{k+1} < c$. The inequality is checked for the actual held binary64 value interpreted as an exact rational. If no nonzero command passes both tests, take $u_k = 0$ and $W_{k+1} = W_k$. This fallback uses the exact reference identity $H'_* = -d_* \le 0$.

Theorem 5. Physical-energy safety over each hold. *Assume the stated guard, uniform error, observation inclusion and reference derivative bounds, and let every actual hold satisfy (50) and its rounded endpoint check. Then*

$$H_*(\tau_k + s) \le \max\{W_k, W_{k+1}\} < c \qquad (0 \le s \le h_k).$$

The conclusion holds throughout every certified finite sequence of holds and preserves the clearance consequence of the energy guard.

Proof. Before first guard exit, integrate (48) using the sampled port and dissipation enclosures to obtain $H_*(\tau_k + s) \le W_k + sB_k + s^2A_k/2$, where $B_k = u_k\widetilde{y}_k + |u_k|\epsilon_{y,k} - \underline{d}_k$ and $A_k = |u_k|L_y + L_d \ge 0$. Since $s^2 \le h_k s$, this upper bound lies below the line joining W_k to $W_k + I_k(u_k)$. Clipping and upward endpoint rounding only raise that line. Both endpoints remain below $c < E_g$, excluding first exit. Zero input is covered by nonincreasing reference energy, and induction proves the result. Supplementary Section S10 gives the complete quantization proof and the required modification when only an upper bound on the actual hold duration is available.

The nominal request uses the same full-phase model. For the output $s = d^{\mathsf{T}}q$, define

$$\widehat{a}_0 = \widehat{H}_{pq}\widehat{v} - \widehat{H}_{pp}(\widehat{H}_q + \widehat{R}\widehat{v}), \qquad \beta = d^{\mathsf{T}}\widehat{H}_{pp}g.$$

When $\beta > 0$, the request is

$$u_{\mathrm{nom}} = \frac{r'' + k_p(r - s) + k_d(r' - d^{\mathsf{T}}\widehat{v}) - d^{\mathsf{T}}\widehat{a}_0}{\beta}.$$

The filter selects an admissible command close to this request. Safety follows from the held-input tests and does not require tracking convergence or inversion of β at every guard state.

Table 5. Certified sampled feedback for the frozen full-phase energy and separately fitted complex damping head. Time and energy are dimensionless.

Quantity		Certified value or bound
Number and length of holds		200, $h = 1/1000$
Total horizon		$\tau \in [0, 1/5]$
Actual input cap		$\lvert u_k\rvert \le 1/10$
Safety energy cap		$c = 3/2000$
Outer guard energy		$E_g = 3003/2000000$
Whole-horizon energy	physical	$H_* < 5.810 \times 10^{-4}$
Planar displacement	centreline	$< 0.777\,\mathrm{mm} < 4\,\mathrm{mm}$

5.3 *Finite-horizon validation on the reference plant*

The numerical instance uses $r(\tau) = 0.003\sin(0.5\tau)$, $k_p = 4$, $k_d = 3$, $\alpha = 1/2$ and $\Delta_W = 2^{-80}$. The initial state is $(q_1, q_2, p_1, p_2) = (1/80, 1/80, 1/200, 1/1000)$. The total horizon is $t = (1/5)/\omega_0 \simeq 17.52$ ms in SI units; the nominal sine request has dimensionless period 4π. Table 5 reports the resulting certified primary trajectory. An independent interval initial-value validation encloses the true reference solution at all 201 samples and over the 200 complete holds. Each canonical observation box has radius 10^{-7}. All 200 actual commands are nonzero and satisfy the cap. The energy-to-geometry barrier bounds the specified planar centreline displacement by 0.777 mm, within the prescribed 4 mm clearance.

A trajectory computed with tighter numerical tolerances provides a consistency diagnostic; it has no second independent initial-value certificate. Agreement between the two numerical trajectories is therefore not the safety argument. The certified instance concerns the specified finite-Gauss, full-cosine reference with declared distributed viscosity, zero unmodelled external force, exact mathematical hold times and the recorded represented commands. The viscosity is a numerical constitutive assumption. Sensor, actuator, delay and physical clock uncertainty require additional budgets. The finite-horizon certificate establishes neither infinite execution nor tracking convergence, and the conservative KAM theorem is not applied to these dissipative closed-loop equations.

6 Discussion

The results establish several linked properties of the actual full-phase, precision-centered GC output. The 736-coordinate complex network is trained with a fixed mechanical prior and a previously fitted PSD head, and the returned

scalar retains configuration and momentum dependence. A realized 129-dimensional observation certificate bounds its complete first- through third-order error derivatives. The same conservative Hamiltonian and its learned–reference family have certified invariant two-tori, while its pairing with the specified dissipative head supports a separate finite-horizon physical-reference energy certificate. Maintaining the identity of these functions is essential: each certificate concerns its own energy, state domain and observed or computed quantities.

The full-phase correction changes the connection between learning and mechanical interpretation. The prescribed kinetic graph still supplies a useful canonical observation chart, but the learned velocity also contains N_p. This derivative appears in the kinematic training loss, the predicted damping force, the output-error certificate and the physical power reserves. A rest-gradient fit alone cannot settle these momentum terms. The combined configuration and momentum design addresses them through structural reference labels and actual mixed-derivative remainders. The resulting deterministic bound supports an inference about the deployed network even though its finite representative is only an auxiliary verification object.

The conservative certificate covers each real member of $H_\lambda = T_G + V_4 + (1-\lambda)N + \lambda U$, $0 \le \lambda \le 1$. The observation-derived derivative bounds enter the uniform field constants for this same family. Its actual varying mean torsion, complete degree-ten embedding prefix, and all angle and parameter tails are retained. The complete defect is below 3.848×10^{-41}, compared with a sufficient budget exceeding 9.303×10^{-40}. Every one of the eight a posteriori quotients is below 0.04136, and the real-embedding test includes the corrected map. Consequently, for every real λ, there exists a real analytic embedded invariant two-torus with the same exact rotation vector, analytic on the strip of width $1/400$. Its distance from the corresponding degree-ten polynomial is uniformly below 3.949×10^{-22}. At $\lambda = 1$ the Hamiltonian is precisely the specified finite-Gauss conservative reference. The conclusion is per parameter with uniform bounds; it makes no assertion of analytic parameter dependence or an attracting torus of the forced or dissipative flow.

The magnetic-feedback certificate instead uses physical energy and power. The PSD decoder imposes a sign on learned unforced power, while the reference-energy ledger accounts for matrix error, momentum-gradient error and input uncertainty within each declared observation box. The validated baseline record covers 200 actual holds and the true reference initial-value problem over its finite horizon. The refined record checks numerical consistency; it does not add a second validated reference trajectory. This evidence establishes the stated energy cap and domain containment. Tracking convergence, arbitrary future execution, and sensor or actuation accuracy require their respective additional analyses.

The experiments give a qualified assessment of complex representations. In the original four-cell comparison, the quadratic prior improves both representations, and the complex cells have smaller rest-gradient errors for the three paired optimization seeds. The five contrasts describe that architecture pair and budget. Their absolute errors remain larger than the zero-correction fourth-order prior's error. Recentring the objective and rescaling the output produce a substantially different comparison: the real prior-assisted network is more accurate, while both repaired outputs still fall short of the shared prior on the same labels. The real damping head also performs better in the reported identification comparison. These observations show that a mathematical certificate for a learned model and an accuracy benefit from learning are separate empirical questions. The GC certificates establish the former; the available rest-gradient comparisons do not establish the latter relative to the fourth-order prior.

Several features explain why a causal comparison of real and complex arithmetic remains unresolved. The networks differ in hidden width, constrained parameter domain and initialization-induced function scale, as well as their small difference in real parameter count. The three factorial seeds repeat optimization on common data. The rest labels and trajectory diagnostics are reused validation, and rest gradients test a configuration slice. Theoretical results for specified complex-reaction architectures (Zhang et al. 2022) do not remove these experimental dependencies or establish a guarantee for the present Adam fits. An informative extension would match initial function scales and resolved signal levels while evaluating full-phase errors on an independent design.

The statistical acceptance result identifies another precise extension. With its stipulated independent-anchor law and common fidelity event, the same good event controls all outputs that obtain sound individual certificates. Data-dependent selection is compatible with this implication, and the false-acceptance probability is bounded by the probability of leaving that event. The current prescribed trajectories instantiate the realized deterministic inverse, rather than this independent sampling experiment. A class-uniform learning guarantee also needs its specified regularity and complexity assumptions (Chen et al. 2022).

Optimization success is a further question: the certificate is evaluated for the returned output without assuming that Adam reaches a global optimum or a stationary point.

The physical scope remains the specified finite-Gauss rod reduction and declared distributed viscosity. The known inertia and elastic prior are shared with the simulator; they are not inferred from an unknown device. Continuum approximation, viscosity calibration and observation fidelity therefore require separate physical evidence. The present Rayleigh decoder is configuration dependent and quadratic in velocity. Velocity-dependent drag, contact and additional environmental forces would change its structural and error-transport requirements. Within the declared model, the combination of finite-output acceptance, conservative verification and physical power accounting makes those requirements explicit and provides a basis for testing stronger models and independent observations.

Data and code availability

The frozen network parameters, native COMSOL observations, exact-rational and interval certificates, and verification sources are indexed in the accompanying reproducibility manifest. The supplementary material contains the full proofs and numerical-certificate conventions.

Declarations

This study reports mathematical analysis and simulated mechanical data. It involved no human participants or animals.

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